T-REX: Twisted Replica Exchange for Bootstrapping Reasoning Guidance in LLMs

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1 Introduction

Reinforcement learning with verifiable rewards (RLVR) is currently the predominant paradigm to improve reasoning capabilities in large language models (LLMs) [Lightman et al., 2024, Shao et al., 2024, DeepSeek-AI, 2025, Lambert et al., 2025]. In RLVR, a pretrained "base" model is fine-tuned using a reward signal derived from a verifier that is meant to judge objective correctness. Repeating this feedback loop is intended to increase the probability that the model produces correct answers and, in turn, to encourage the emergence of more effective reasoning behaviors.

However, recent results have called into question whether RLVR reliably induces qualitatively new reasoning. Yue et al. [2025] find that RL does not elicit new reasoning paths inaccessible to the base model, while Karan and Du [2025] show that sharpening the base model's sampling distribution can match the performance of RL-finetuned models across several reasoning benchmarks. Furthermore, RL may also degrade the diversity of LLM-generated outputs, exhibiting the "mode collapse" phenomenon [Jiang et al., 2025].

Motivated by these concerns, a growing line of work reframes inference-time scaling as approximate probabilistic inference [Levine, 2018, Lew et al., 2023, Zhao et al., 2024, Feng et al., 2025, Puri et al., 2025]. In probabilistic inference, we characterize a target (posterior) distribution we are interested in, and approximate it by attempting to sample from it and estimating a normalizing constant. In the context of LLMs, the target distribution is often the base model distribution reweighted by constraints such as verifiable correctness. Then, reasoning in LLMs can be viewed as traversing a sequence of intermediate latent variables (e.g., reasoning tokens/steps used as partial solutions or plans) that guide sampling toward correct outputs. Providing an approximation of the full posterior has certain beneficial properties, such as the preservation of multiple modes of the target distribution, tackling the mode-collapse problem previously mentioned.

The "narrow passage" problem in sampling describes the challenge of sampling from a distribution where the probability mass is concentrated in a tiny, isolated region of the state space [Orthey and Toussaint, 2021, Li and Dantam, 2023, Li et al., 2025, Gundersen and Cornish, 2025]. In the context of LLMs, this phenomenon is acute during complex reasoning tasks: a single logical error or arithmetic mistake in an early step can render the entire subsequent trajectory invalid. While LLMs excel at intuitive generation, they struggle with these hard constraints because standard autoregressive sampling is myopic; it cannot see the future constraint violation until it has already occurred.

[NOTE: Different reasoning problems are going to have the solution's probability mass concentrated in very different areas of the state space in the extremely large and intractable space of output sequences. My opinion is that we essentially need an automated way to, at test time, find and efficiently sample that region of the state space for each problem we are facing, and have that sample survive generation until the end so that we can at least receive some reward signal from verifying the solution.]

[NOTE: I will make some assumptions.
(1) The base model puts some probability on the "correct" region of state space, although the likelihood of this region under the model can be (but is not always) lower than the model main modes. Because the solution's likelihood may be lower than other regions of state space, sharpening the base model distribution can kill off the chances of sampling a correct answer or correct reasoning step.

(2) If the model does sample a correct solution, the verifier is powerful and robust enough to recognize the correctness of the answer even though the reasoning might be unorthodox or not of high likelihood in the base model.]

Given these assumptions, the problem is that we would like to, at test time, provide some kind of learned twist or critic that would allow us to find and efficiently sample that region of the state space for each reasoning problem that the model receives. I plan to use the following methods:

1. **Exploration-Based (Parallel Tempering):** These methods run parallel chains at varying temperatures to flatten the energy landscape, allowing the model to traverse barriers between modes. However, in the high-dimensional discrete state space of text, heating a chain often results in a diffusive random walk. Without direction, the model wastes compute exploring incoherent regions of the state space rather than semantically diverse reasoning paths.

2. **Guidance-Based (Twisted SMC):** These methods learn a "twist" (or value) function to bias the token proposal distribution toward high-value regions, effectively steering the model through the narrow passage. However, these methods face a fundamental "chicken-and-egg" paradox: to learn an accurate twist function, one needs a dataset of valid solution trajectories; yet, these valid trajectories are initially unknown and we have no method to sample them efficiently.

We propose T-REX, a framework that unifies these approaches to resolve their respective limitations. We leverage the diversity of Replica Exchange to break the "chicken-and-egg" cycle of Twisted SMC.

In reasoning tasks, heating the model slightly would improve semantic diversity whilst not producing random noise. It would increase the probability mass of the model over the "correct" solution region of the state space, given the assumption that we already have a non-negligible probability over the solution. This would allows the hot chains to brainstorm alternative reasoning strategies (e.g., algebraic vs. heuristic) via the model's strong prior. These hot chains discover the initial sparse solutions required to train the twist function online. In turn, the learned twist guides the cold chains, progressively narrowing the search space around valid solutions.

This approach establishes a new paradigm for Adaptive Test-Time Compute. Rather than treating the proposal distribution as static, we demonstrate that inference compute should be allocated dynamically: not just to search for solutions, but to learn how to search in real-time. By coupling global exploration (Parallel Tempering) with local guidance (Twisted SMC), we enable LLMs to reason through narrow passages that are inaccessible to standard sampling methods.

2 Methodology (need to update this section)

2.1 Problem Formulation

[NOTE: Instead of framing it as reasoning steps, we might go back to the token-level twist formulation. Reason being that if we resample at reasoning step boundaries, we face the issue of parallelism; some particles may have longer reasoning steps than others and have to wait for all to finish before resampling, and it may also lead to issues with training signal for twist functions]

We are given a prompt x_0 containing a problem statement, and generate a solution, which is a sequence $x_{1:T}$, from a base LLM (prior) P_θ. For simplicity, we always assume that $x_{1:T} \sim P_\theta(\cdot \mid x_0)$ and omit the dependence on x_0 for ease of notation. The solution $x_{1:T}$ can be decomposed into variable-length reasoning steps x_i, and the LLM generates all steps autoregressively. Each solution contains the reasoning process and answer to the problem. The solution $x_{1:T}$ is subject to strict, non-local constraints $\phi(x_{1:T}) \in \{0, 1\}$ where the value of ϕ is 1 if the solution $x_{1:T}$ satisfies all constraints, and 0 otherwise. The target distribution is the posterior:

$$\pi(x_{1:T}) = \frac{1}{\mathcal{Z}_\pi}\tilde{\pi}(x_{1:T}) = \frac{1}{\mathcal{Z}_\pi}P_\theta(x_{1:T})\phi(x_{1:T}) \propto P_\theta(x_{1:T})\phi(x_{1:T}) \tag{1}$$

where $\tilde{\pi}(x_{1:T}) = P_\theta(x_{1:T})\phi(x_{1:T})$ is the unnormalized target density, and $\mathcal{Z}_\pi = \sum_{x_{1:T}} \tilde{\pi}(x_{1:T})$ is the intractable normalizing constant. The set of valid reasoning trajectories constitutes a sparse subspace within the vast combinatorial output space of the prior P_θ.

2.2 From Importance Sampling to Twisted Sequential Monte Carlo

Importance Sampling. Importance sampling-based methods [Tokdar and Kass, 2010] attempt to estimate the normalizing constant by sampling from a proposal or importance distribution q, and reweighting samples by the ratio of the unnormalized target and proposal densities. In a set of samples, the *importance weight* of the i'th sample is:

$$w(x_{1:T}^{(i)}) = \frac{\tilde{\pi}(x_{1:T}^{(i)})}{q(x_{1:T}^{(i)})}. \tag{2}$$

Using a proposal distribution that can be easily sampled from, importance weights allow for the estimation of expectations under the normalized target distribution π of any arbitrary given function using the Monte Carlo method. However, the variance of the importance weights significantly affects the quality of the Monte Carlo estimate. High weight variance leads to weight degeneracy; when a few samples with high weights dominate the simulation, the effective sample size (ESS) collapses. Setting the proposal to equal the target makes the importance weights have zero variance, but in practice, we cannot do this.

Sequential Monte Carlo. Sequential Monte Carlo (SMC) addresses degeneracy by replacing one hard importance sampling step with a sequence of unnormalized intermediate target distributions $\{\tilde{\pi}_t(x_{1:t})\}_{t=1}^{T}$ that are easier to sample, where $\pi_t = \frac{\tilde{\pi}_t}{\mathcal{Z}_t}$. SMC maintains N sequences (each sequence is also called a particle), and at each step t, the following procedure is carried out:

1. **Propagate:** For each sequence $x^{(i)}$ where $i \in 1, \ldots, N$, sample the next reasoning step:

$$x_t^{(i)} \sim q(x_t \mid x_{1:t-1}^{(i)}) \tag{3}$$

2. **Reweight:** Compute the *incremental importance weight* for each sequence based on the intermediate target:

$$w_t(x_{1:t}) = \frac{\tilde{\pi}_t(x_{1:t})}{\tilde{\pi}_{t-1}(x_{1:t-1})q(x_t \mid x_{1:t-1})} \tag{4}$$

3. **Resample:** Once all sequences have their incremental importance weights, resample N particles with replacement from the existing N particles proportional to their respective incremental importance weights.

SMC avoids the problem of weight degeneracy because low potential sequences are quickly washed out in the resampling step in favour of high potential sequences. This allows for greater local exploration that better covers the high-probability areas of the target distribution that are covered by a given prefix that is duplicated more than once. SMC also has the benefit of providing an unbiased estimator of the final target normalizing constant:

$$\hat{\mathcal{Z}}_\pi = \prod_{t=1}^{T} \frac{1}{N} \sum_{i=1}^{N} w_t(x_{1:t}^{(i)}) \tag{5}$$

Twisted Sequential Monte Carlo. The spirit of Twisted Sequential Monte Carlo (TSMC) is in how the intermediate target distributions are constructed. Given that the only constraint we are interested in is that the final (intermediate) target distribution π_T matches the final target we are interested in, i.e. $\pi_T = \pi$, then we are free to construct the intermediate targets as we like, and can therefore construct them so that the incremental importance weights we compute allow a higher number of partial prefixes that are going to lead to mass in the final target to survive resampling.

[TODO: describe twisted SMC here.]
[NOTE: The early part of the sequence generation is the most important since that locks us into a general reasoning direction/strategy, so we want to try and have as diverse of a starting point as possible...]
[NOTE: Setting the proposal to the base model multiplied by the twist could make the optimal proposal intractable if we are resampling after a select number of tokens (e.g. 8-64 tokens). If we resample every single step, we face the difficulty of compute explosion (too many twist functions) and also with short extensions, the extra value added or derived from adding just one or a few tokens is not much, too noisy.]

[NOTE: Resampling at reasoning steps can also be very bad as mentioned earlier. We really need to think about what is the right way to have twists integrate into this framework, and what can be a smarter proposal than simply just using the base model. We want to have that balance of exploration with the parallel tempering hot temperatures and exploitation with the twist guided proposals.]

2.3 Why Tempering? (Smoothing the Landscape)

To traverse this challenging state space, we employ Parallel Tempering (PT) [Swendsen and Wang, 1986, Geyer, 1991]. We construct a sequence of K distributions $\{\pi^k\}_{k=1}^K$ defined by an inverse temperature schedule $0 = \beta_1 < \cdots < \beta_K = 1$:

- **At $\beta \approx 0$ (Hot):** The landscape is flat. Barriers are removed, allowing hot particles to traverse the space freely and discover diverse (though potentially invalid) proto-solutions.
- **At $\beta \approx 1$ (Cold):** The landscape is rugged. Cold particles are locked into valid regions but cannot cross energy barriers to find better modes.

By exchanging states between these temperatures, we allow cold chains to escape local optima by teleporting to a hotter state, moving across the barrier, and cooling back down in a new valid mode.

2.4 Global Communication: Non-Reversible (Discrete) Transport

Standard PT fails in high dimensions (like LLM latent spaces) for two reasons: (1) Random walk behavior makes traversing the temperature ladder slow ($O(K^2)$ steps), and (2) Lack of distributional overlap makes swap acceptance rates vanish. We address (1) by using Non-Reversible Parallel Tempering (NRPT) [Syed et al., 2022]. We will have to see how to address (2); this could be a challenging problem in practice. [NOTE: We can maybe plug-and-play various transport mechanisms from simpler to more complicated and see what sticks, but perhaps what I envision which would be cooler is a transport mechanism based on Block-Gibbs.]

2.4.1 Non-Reversible Parallel Tempering (NRPT)

To solve the random walk problem, we discard the standard practice of choosing swap pairs at random. Instead, we use a deterministic, alternating schedule [Syed et al., 2022]:

$$\mathcal{S}_{odd} = \{(1,2),(3,4),\dots\}, \quad \mathcal{S}_{even} = \{(2,3),(4,5),\dots\} \tag{6}$$

By toggling between these sets at even/odd timesteps, we induce a ballistic flow. A successful particle will propagate from the prior to the posterior in $O(K)$ steps, significantly faster than the diffusive $O(K^2)$ of standard PT.

2.4.2 Transporting Particles Between Temperatures (e.g. Adaptive Block-Gibbs)

LLM latent spaces and text sequences exist in extremely high-dimensional, discrete state spaces. In high dimensions, it is likely that if the temperatures are too different, the distributions of the hot and cold chains have virtually no overlap. The probability of accepting a swap between two temperatures vanishes to almost zero. However, having too many temperatures so that the tempered distributions are close enough that they overlap might be computationally very expensive and slow. [NOTE: I don't know much about transport. My understanding is that transport is essentially a way to align the distributions of the different temperature chains.]

Although neural transports are a natural way to facilitate moves between temperatures [Zhang et al., 2025], continuous normalizing flows are inapplicable to discrete token sequences due to the discretization bottleneck.

My initial idea is some kind of Adaptive Block-Gibbs Transport. Instead of learning a global transport map, we learn a local Discrete Editor that modifies a "hot" particle $x^{(k)}$ to increase its likelihood of acceptance in the "cold" distribution π^{k+1}. We decompose this transport into two phases: *Learned Masking* and *Base In-filling*.

Phase 1: The Learned Masking Policy We employ a lightweight Critic network parameterized as a token-level binary classifier. Given a trajectory x and a target temperature β, the Critic predicts

a binary mask $\mathcal{M} \in \{0,1\}^T$, identifying tokens responsible for constraint violations (e.g., logical fallacies or syntax errors): The policy can be conditioned on β, allowing the editor to be lenient at high temperatures (masking only syntax) and strict at low temperatures (masking subtle logic errors). [NOTE: we could distill this from a trained PRM. The reason why we wouldn't want to use a large PRM is because inference would be much slower, but also the proposal reversibility would be broken since the PRM would almost certainly produce a different mask $\mathcal{M}$ for the two different particles x and x']

Phase 2: Base Model In-filling To generate the repair, we freeze the unmasked context $x_{\setminus\mathcal{M}}$ and resample the masked tokens using the **base pre-trained LLM** P_θ:

$$x'_{\mathcal{M}} \sim P_\theta(x_{\mathcal{M}} \mid x_{\setminus\mathcal{M}}) \tag{7}$$

Crucially, we use the base model P_θ rather than the twisted proposal for this step. This ensures that the proposal kernel matches the prior, significantly simplifying the acceptance ratio (see below). The repair is generated autoregressively to completion before verification.

The Transport Acceptance Ratio We treat the edit $x \to x'$ as a Metropolis-within-Gibbs proposal. The acceptance probability α for swapping the transported particle x' into chain $k+1$ is:

$$\alpha = \min\left(1, \frac{\pi_{k+1}(x')}{\pi_{k+1}(x)} \times \frac{q(x|x')}{q(x'|x)}\right) \tag{8}$$

By substituting the proposal kernel $q(x'|x) = \pi_\phi(\mathcal{M}|x)P_\theta(x'_{\mathcal{M}}|x_{\setminus\mathcal{M}})$ and the target density $\pi(x) \propto P_\theta(x)\psi^*(x)^\beta$, the complex language model likelihoods $P_\theta(\cdot)$ cancel out. The ratio reduces to:

$$\alpha = \min\left(1, \underbrace{\frac{\psi^*(x')^{\beta_{k+1}}}{\psi^*(x)^{\beta_{k+1}}}}_{\text{Constraint Gain}} \times \underbrace{\frac{\pi_\phi(\mathcal{M}|x')}{\pi_\phi(\mathcal{M}|x)}}_{\text{Critic Symmetry}}\right) \tag{9}$$

This formulation makes the transport computationally tractable, as it avoids calculating the intractable normalization constant of a twisted distribution.

3 Related Work

there might be other works to mention...

Discrete Feynman-Kac Correctors?

3.1 Reinforcement Learning and Control as Inference

Our work builds upon the theoretical equivalence between optimal control and probabilistic inference, where maximizing expected reward is dual to sampling from an unnormalized target distribution [Levine, 2018]. In the context of Large Language Models (LLMs), alignment techniques such as RLHF using PPO [Schulman et al., 2017] or GRPO [Shao et al., 2024] can be viewed as variational inference methods that attempt to fit a parameterized policy π_θ to a target distribution typically defined by a reward function [Zhou et al., 2025]. However, RL approaches may fail to elicit new reasoning paths that the base model cannot generate, and since they sharpen the output distribution of the model, may lead to mode collapse, worsen generation diversity, and worsen generalization to novel logical constraints at test time [Yue et al., 2025, Karan and Du, 2025].

By utilizing high-temperature chains to explore the state space, T-REX implements a form of entropy-regularized exploration [Haarnoja et al., 2017], collecting off-policy trajectories to bootstrap the training of the twist (critic) and guide the zero-shot inference of the base model.

3.2 Inference-Time Scaling via Probabilistic Inference

Recent research has re-framed inference-time scaling from a search problem (e.g., Tree Search, Best-of-N) to a formal probabilistic inference task. This trend spans multiple generative modalities.

In the diffusion domain, Dou and Song [2024] formulated inverse problem solving as non-linear filtering, demonstrating that Sequential Monte Carlo (SMC) can rigorously enforce constraints that heuristic guidance cannot. Similarly for Large Language Models (LLMs), Puri et al. [2025] modeled text generation as a state-space model, showing that Particle Filtering significantly outperforms search-based baselines like Beam Search on mathematical reasoning benchmarks. However, these standard SMC approaches rely on the pre-trained model for proposals ($x_t \sim P_\theta(\cdot|x_{t-1})$), utilizing the reward model only for re-weighting. This leaves them vulnerable to the narrow passage problem in hard-constrained settings, where the standard prior rarely proposes valid tokens to be re-weighted.

3.3 Twisted Sequential Monte Carlo

To address the inefficiency of blind proposals, Twisted SMC methods [Heng et al., 2020] introduce a learned auxiliary function (the twist) to guide the proposal distribution toward high-value regions. In the context of language models, Zhao et al. [2024] and Feng et al. [2025] have applied Twisted SMC to steer generation. Feng et al. [2025] learn a value function V_ϕ to tilt the proposal $q(x_t) \propto P_\theta(x_t)e^{V_\phi(x_t)}$, while Zhao et al. [2024] learn to minimize the KL-divergences between the intermediate target marginals and the twisted intermediate target distributions.

While effective, these methods typically suffer from a Catch-22: learning an accurate twist function requires high-quality trajectories of valid solutions, which are initially unknown and difficult to sample without an already-trained twist. Most recently, Kim et al. [2025] addressed this via *Self-Distilled Twisted SMC*, which iteratively refines the base model via self-distillation to progressively align it with the target. By generating samples with the current model and retraining on them, they generate the necessary training data to learn the twist over multiple generations. However, this approach relies on an expensive, multi-stage iterative retraining loop rather than a single inference pass.

3.4 Parallel Tempering and Replica Exchange

Parallel Tempering (Replica Exchange) [Swendsen and Wang, 1986, Geyer, 1991] is a Markov Chain Monte Carlo (MCMC) method designed to overcome mode collapse by running multiple chains in parallel at varying temperatures. Recently, this framework has been successfully adapted to control generative models and enhance sequential inference.

In Continuous Sampling and Generative Control: Recent work has demonstrated that learning-based transport can dramatically accelerate Parallel Tempering. Zhang et al. [2025] introduced *Accelerated Parallel Tempering (APT)*, which employs Neural Transports (such as Normalizing Flows and Diffusion Models) to warp the state space between temperature ladders. By learning a continuous transport map, APT minimizes the distributional mismatch between chains, thereby maximizing the swap acceptance rate even between disparate distributions.

Similarly, in the domain of diffusion models, the CREPE algorithm applies Replica Exchange to inference-time control [He et al., 2025]. By running parallel diffusion chains at different noise levels and exchanging states based on a constraint reward, CREPE steers generation towards user-defined constraints while maintaining superior mode coverage compared to standard SMC-based debiasing methods.

In Sequential Settings: The principles of Replica Exchange have also been extended to sequential inference tasks. Shestopaloff and Doucet [2019] proposed using parallel replicas to provide lookahead information for Sequential Monte Carlo (SMC), conditioning the current particle's proposal on the future paths of other replicas to mitigate myopia. More recently, Nabika et al. [2025] proposed *Sequential Exchange Monte Carlo (SEMC)*, which integrates SMC with Replica Exchange to automate the tuning of temperature schedules and step sizes during the annealing process.

4 Proposed Experiments

A lot of the experimental work should build off choices made by prior work, e.g. reasoning for math datasets used in Lightman et al. [2024], Feng et al. [2025], Lambert et al. [2025], Yue et al. [2025], Karan and Du [2025].

4.1 Models and Benchmarks

We will do experiments with Qwen2.5 (7B/14B/32B base variants) [Qwen et al., 2025] and LLaMA-3.1-8B [Grattafiori et al., 2024].

Use stratified sampling to reduce variance.

My pipeline for now is to first do supervised fine-tuning on the base model on a filtered subset of PRM800K [Lightman et al., 2024] following the procedure in Feng et al. [2025], which allows the base model to learn step-by-step reasoning. After that, we would train the twist model, and then test.

We focus on mathematical reasoning tasks:

- **GSM8K [Cobbe et al., 2021]:** Where an error in step 1 invalidates the final answer.
- MATH or MATH500 [Hendrycks et al., 2021]; more complicated math questions. Sometimes, only a subset of 500 questions are used for eval.
- If we need more challenging math benchmarks, consider AMC or AIME benchmarks.
- HumanEval: programming
- GPQA, GPQA Diamond: Science multiple choice questions

It might be interesting to see if we can extend to other tasks like this narrow passage problem, e.g.

- **Code Generation**
- **Molecule Generation (protein language models)**

4.2 Baselines

- **Best-of-N brute force rejection sampling:** defines the difficulty of the problem for the base model. Clone yourself N times, each clone attempts the problem. You then look at all N distinct solutions. If even one clone gets the right answer, you count the problem as solved. Run this baseline at different temperatures to assuage reviewers concerns: $[0.6, 0.8, 1.0, 1.2]$. We could take the first 100 problems and do a sweep on these temperatures before picking one temperature for the final run on all benchmark problems.
  - Defines the difficulty of the problem: how many samples are needed to get a valid answer? 100? 10,000?
  - Efficiency: is our method able to find the answer faster (less time) and with fewer samples? Our method learns to search the space more efficiently...
- **PPO and GRPO:** With KL-to-prior penalty $\beta = 0.01 \ldots 0.05$.
- **Standard SMC Steering with a Process Reward Model:** work by Puri et al. [2025].
- **Twisted SMC using the base model as proposal and only using twists in weights for resampling:** i.e. Feng et al. [2025]
- **SMC with multiple temperatures:** Parallel tempering SMC without twists. Tests the value of the twists.

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